

NAME

Tsat – compute the saturation pressure

LIBRARY

iapws library (*libiapws*, *-liapws*)

SYNOPSIS

```
Fortran      pure subroutine Tsat(ps,Ts)
              real(dp), intent(in), contiguous :: ps(:)
              real(dp), intent(out), contiguous :: Ts(:)

C            void iapws_r797_Tsat(size_t N,double *ps,double *Ts)

Python      Tsat(ps:np.ndarray)->Union[np.ndarray,float]:
```

DESCRIPTION

Arguments:

ps(:) Saturation pressure in MPa.

Ts(:) Saturation temperature in K.

Validity range: 611.213 Pa <= *ps* <= 22.064 MPa

RETURN VALUE

Ts is filled with NaN if out of validity range.

EXAMPLES

Fortran:

```
use iapws
real(dp) :: T(1), p(1)
p(1) = 0.1_dp
call Tsat(p, T)
```

C:

```
#include "iapws.h"
double p = 0.1;
double T;
iapws_g704_Tsat(1, &p, &T);
```

Python:

```
import numpy as np
import pyiapws
p = np.asarray([0.1])
T = pyiapws.Tsat(p)
print(T)
```

SEE ALSO

iapws(1) **iapws_lib**(3) **iapws_psat**(3) **iapws_Tsat**(3) **iapws_kh**(3) **iapws_kd**(3) **iapws_kw**(3) **iapws_Tc**(3)
iapws_pc(3) **ciaaw**(3), **codata**(3),